# Discord Math Server & Shark Math Program Mathematics Competition

## 2023 Round 1

a person, Daniel, mathemolympiad,
RDES, yoavmal

July 2023

## Instructions and Information

1. DO NOT OPEN THE BOOKLET UNTIL YOUR CONTEST TIMER STARTS.
2. Teams can consist of up to 4 people. Each team should sign up before Round 1.
3. This is a 19-question, 2-day test. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
4. No calculators, books, online resources, etc. are allowed, and no other tab aside from the test paper should be opened. Only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
5. No person who participated or helped in the contest is allowed to discuss any aspect of it 48 hours from the contest date. If moderators catch anyone discussing during this time frame, they will automatically be disqualified.
6. The cutoff score is the maximum score such that at least 10 teams move on to the second round.

*Good luck to all contestants and enjoy the contest!*

## Problems

1. **[1]** Find the sum of the first 20 even integers.

2. **[1]** When flipping a fair coin 6 times, the probability of flipping at most 2 heads is $\frac{m}{n}$ where $m$ and $n$ are relatively prime positive integers. find $m+n$.

3. **[1]** If $\overline{ABABA}$ is a 5-digit number divisible by 72, where $A, B$ are single digit numbers ($A \neq 0$), find the sum of all possible values of $\overline{AB}$.

4. **[1]** A sphere contains a cube inside it such that the sphere circumscribes the cube. If the volume of the cube is 64, then the volume of the sphere can be expressed as $a\pi\sqrt{b}$ where $a$ and $b$ are positive integers, with $b$ not being divisible by the square of any prime. Find $a+b$.

5. **[2]** In a school, the ratio of the number of boys to the number of girls, the number of boys who wear glasses to the number of girls who wear glasses, the number of boys who do not wear glasses to the number of girls who do not wear glasses, and the number of students who wear glasses to the number of students who do not wear glasses are $4:5, 2:3, 4:3$ and $x:y$ respectively where $x$ and $y$ are relatively prime positive integers. Find $99x+6y$.

6. **[2]** A point $P$ is in the interior of equilateral $\triangle ABC$. It is known that the perpendicular distance from $P$ to $AB, BC, CA$ are $2, 3, 4$ respectively. Find the square of the area of $\triangle ABC$.

7. **[2]** Let $a = \sin 2023°$ and $b = \cos 2023°$. Then, $(a+b)^2$ can be written as $1 + \sin x°$. Find the minimum positive value of $x$.

8. **[2]** A real number $x$ satisfies $\log_{128}(\log_{81} x) = 22$. Evaluate $\log_2(\log_3 x)$.

9. **[2]** The minimum value of $x - 3 + \frac{10}{x+3}$ for real $x > 0$ can be expressed as $a\sqrt{b} + c$ for integers $a, b, c$ such that $b$ is a positive integer not divisible by the square of any prime. Find $a+b+c$.

10. **[2]** Points $A(0,4), B(5,0), C(5,-4)$ lie on the Cartesian Coordinate Plane. When $D(x,y)$ is added to the plane, $A, B, C, D$ form a non-self intersecting trapezium with its vertices being in this order, such that the area of the trapezium is 28. If the sum of all $x+2y$ among all possible coordinates of $D$ is $\frac{m}{n}$ where $m$ and $n$ are relatively prime integers with $n > 0$, find the value of $100m+n$.

11. **[3]** If the product of all solutions to
$$x^2 = 2\left(2|x-1| + x - 1\right)$$
is $n$, find $\lfloor n \rfloor$.

12. **[3]** In a $2022 \times 2022$ grid, we write a number in each cell, such that for any rectangular sub-grid which includes the top left cell, the average of the numbers in the sub-grid is equal to the number of cells in the sub-grid. How many of the cells in the grid contain a number divisible by 9?

13. **[3]** How many positive divisors of $69^{420}$ are there with units digit 7?

14. **[4]** Culver colours the first 16 positive integers in red or blue such that 4 of them coloured in red and the other 12 are coloured in blue. Let $n$ be the number of ways that Culver can arrange the 16 integers from left to right, such that for any two integers of different colour, the larger one must be right (not necessarily directly) of the smaller one. Find the minimum possible value of $n$.

15. **[4]** $\triangle ABC$ is a right-angled triangle with $AB = 12, AC = 16$ and $\angle BAC = 90°$. $DEFG$ is a rectangle with $D$ on $AB$, $E$ on $AC$, and $F, G$ on $BC$. $H$ is a point on the same side as $C$ with respect to line $EF$ such that $BH = CD$ and $\angle HEF = \angle EBC$. If $EH = 7$, find $DE$.

16. **[4]** If $a + \frac{1}{b} = 2, b + \frac{1}{c} = 3, c + \frac{1}{d} = 4, d + \frac{1}{a} = 5$, find $abcd + \frac{1}{abcd}$.

17. **[4]** If $n$ is a positive integer not greater than 100, such that there exist positive integers $a, b$ satisfying
$$6a^2 - 4b^2 = 200^n,$$
find the sum of all possible values of $n$.

18. **[4]** $ABCD$ is a square inscribed in circle $\Gamma$. $E$ is a point on minor arc $BC$ of $\Gamma$. $DE$ intersects $AC$ and $BC$ at $F$ and $G$ respectively. It is known that $FG = 25$ and $GE = 7$. If $DF = \frac{m}{n}$ where $m, n$ are relatively prime positive integers, find $100m + n$.

19. **[5]** A sequence is called *delta* if it consists of 1000 integers $a_1, a_2, ..., a_{1000}$ ranging from 1 to 1000, and there exists no integers $1 \leq i \leq j \leq 1000$ such that $a_i + a_{i+1} + a_{i+2} + ... + a_j$ is divisible by 1001. If Culver randomly chooses a *delta* sequence, the expected sum of its elements is $\frac{m}{n}$, where $m, n$ are relatively prime positive integers. Find $m + n$.

# Solutions

1. Since the first and 20-th positive even integers are 2 and 40 respectively, Sum $= \frac{(2+40)\cdot 20}{2} = \boxed{420}$.

2. The probability is
$$\frac{\binom{6}{0}+\binom{6}{1}+\binom{6}{2}}{2^6} = \frac{1+6+15}{64} = \frac{11}{32}$$
so the answer is $11+32 = \boxed{43}$.

3. As the number is divisible by 3, $3A+2B$ is divisible by 3 i.e. $B$ is divisible by 3. So $B = 0, 3, 6$ or 9.
If $B = 0$, as the number is divisible by 8, $\overline{A0A} = 101A$ is divisible by 8 i.e. $A = 8$. But 80808 is not divisible by 9 so there are no solutions in this case.
If $B = 3$, as the number is divisible by 8, $\overline{A3A} = 101A + 3$. We can easily see that $A = 2$ but 23232 is not divisible by 9 so there are no solutions in this case.
If $B = 6$, as the number is divisible by 8, $\overline{A6A} = 101A + 6$. We can easily see that $A = 4$ but 46464 is not divisible by 9 so there are no solutions in this case.
If $B = 9$, as the number is divisible by 8, $\overline{A9A} = 101A + 9$. We can easily see that $A = 6$. As 69696 is divisible by 72 we have a possible value of $\overline{AB}$ is 69.
Combining all above cases, the answer is $\boxed{69}$.

4. The edge length of the cube is $\sqrt[3]{64} = 4$. So, its space diagonal is $\sqrt{3(4^2)} = 4\sqrt{3}$. Since the sphere circumscribes the cube, the radius of the sphere is $\frac{4\sqrt{3}}{2} = 2\sqrt{3}$. Therefore, the volume of the sphere is $\frac{4\pi}{3}(2\sqrt{3})^3 = \frac{4\pi}{3}(24\sqrt{3}) = 32\pi\sqrt{3}$, hence the answer is $32+3 = \boxed{35}$.

5. Let $xk$ and $yk$ be the number of students who wear glasses and the number of students who do not wear glasses respectively. So there are $\frac{2}{5}xk$ boys who wear glasses and $\frac{4}{7}xk$ boys who do not wear glasses. On the other hand, there are $\frac{4}{9}(x+y)k$ boys. So
$$\begin{aligned}\frac{2}{5}xk + \frac{4}{7}yk &= \frac{4}{9}(x+y)k\\ \frac{8}{63}yk &= \frac{2}{45}xk\\ \frac{x}{y} &= \frac{\frac{8}{63}k}{\frac{2}{45}k}\\ x:y &= 20:7\end{aligned}$$
The answer is hence $99(20)+6(7) = \boxed{2022}$.

6. Let $s$ and $h$ be the side length and the length of height of $\triangle ABC$. By considering areas, we have $\frac{1}{2}(2)(s) + \frac{1}{2}(3)(s) + \frac{1}{2}(4)(s) = \frac{1}{2}(h)(s)$ which gives $h = 9$. So $s = \frac{2}{\sqrt{3}}h = 6\sqrt{3}$. Therefore, the area of $\triangle ABC$ is $\frac{1}{2}(9)\left(6\sqrt{3}\right) = 27\sqrt{3}$ so the answer is $\left(27\sqrt{3}\right)^2 = \boxed{2187}$.

7. One of the Pythagorean trigonometric identities states that $1 = \sin^2\theta + \cos^2\theta$ and one of the double angle formulas states that $\sin(2\theta) = 2\sin\theta\cos\theta$. Therefore, $(a+b)^2 = a^2+b^2+2ab = 1+\sin(2\cdot 2023^\circ) = 1+\sin 4046^\circ = 1+\sin 86^\circ$. Hence, $x = \boxed{86}$.

8. Note that $\log_{a^b} c = \frac{\log_a c}{b}$. So we have

$$\begin{aligned}\log_{128}(\log_{81} x) &= \frac{1}{7}\left(\log_2\left(\frac{1}{4}\log_3 x\right)\right)\\ &= \frac{1}{7}\left(\log_2\left(\log_3 x\right) + \log_2\left(\frac{1}{4}\right)\right)\\ &= \frac{1}{7}\left(\log_2\left(\log_3 x\right) - 2\right)\end{aligned}$$

and hence $\log_2(\log_3(x)) = \boxed{156}$.

9. We apply the AM-GM inequality on $(x+3) + \frac{10}{x+3}$, treating $x+3$ as one term, to get
$$\frac{(x+3) + \frac{10}{x+3}}{2} \geq 2\sqrt{(x+3)\cdot\frac{10}{x+3}} = \sqrt{10},$$
thus $(x+3) + \frac{10}{x+3} \geq 2\sqrt{10}$. Therefore, $x - 3 + \frac{10}{x+3} \geq 2\sqrt{10} - 6$, thus the answer is $2 + 10 - 6 = \boxed{6}$.

10. We split into two cases:
Case 1: $AD//BC$
Then $x = 0$ and $\frac{(4-y)+(0-(-4))(5-0)}{2} = 28$ i.e. $D = (0, -\frac{16}{5})$ which corresponds to $x + 2y = -\frac{32}{5}$.
Case 2: $AB//CD$
We cut the trapezium into parallelogram $ABCO$ and $\triangle AOD$ where $O$ is the origin.
As the area of $ABCO$ is $(5-0)(4-0) = 20$ so the area of $\triangle AOD$ is $28 - 208$. So $\frac{1}{2}(0-x)(4-0) = 8$ i.e. $x = -4$.
As $AB//CD$,

$$\begin{aligned}m_{AB} &= m_{CD}\\ \frac{4-0}{0-5} &= \frac{(-4)-y}{5-(-4)}\\ y &= \frac{16}{5}\end{aligned}$$

So $x + 2y = \frac{12}{5}$.
Combining both cases we have the sum of all $x + 2y$ being $-\frac{32}{5} + \frac{12}{5} = \frac{-4}{1}$ i.e. the answer is $100(-4) + 1 = \boxed{-399}$.

11. We can rearrange as

$$\begin{aligned}x^2 &= 4|x-1| + 2x - 2\\ x^2 - 2x - 4|x-1| &= -2\\ (x-1)^2 - 4|x-1| + 4 &= 3\\ (|x-1| - 2)^2 &= 3\\ |x-1| &= 2 \pm \sqrt{3}\end{aligned}$$

If $|x-1| = 2 + \sqrt{3}$, then $x - 1 = 2 + \sqrt{3}$ or $x - 1 = -2 - \sqrt{3}$ i.e. $x = 3 + \sqrt{3}$ or $-1 - \sqrt{3}$.
If $|x-1| = 2 - \sqrt{3}$, then $x - 1 = 2 - \sqrt{3}$ or $x - 1 = -2 + \sqrt{3}$ i.e. $x = 3 - \sqrt{3}$ or $-1 + \sqrt{3}$. Hence $n = \left(3+\sqrt{3}\right)\left(-1-\sqrt{3}\right)\left(3-\sqrt{3}\right)\left(-1+\sqrt{3}\right) =$

$\left(3^2 - \left(\sqrt{3}\right)^2\right)\left((-1)^2 - \left(\sqrt{3}\right)^2\right) = 6 \cdot (-4) = -24$, hence the answer is $\lfloor -24 \rfloor = \boxed{-24}$.

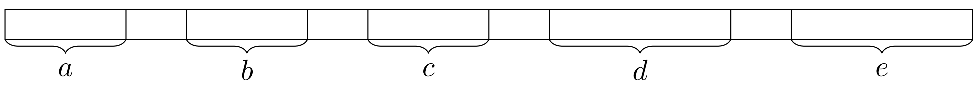
12. Let $(m, n)$ refer to the cell on the $m$-th row and $n$-th column, and $f(m, n)$ be the sum of all numbers in the rectangle with top-left cell $(1, 1)$ and right-bottom $(m, n)$. By the problem condition $f(m, n) = m^2n^2$. As the number in the cell $(m, n)$ is

$$\begin{aligned} &f(m,n) - f(m-1,n) - f(m,n-1) + f(m-1,n-1) \\ &= \left(m^2 - (m-1)^2\right)\left(n^2 - (n-1)^2\right) \\ &= (2m-1)(2n-1) \end{aligned}$$

Hence either one of $2m - 1$ or $2n - 1$ is divisible by 9, or they are both divisible by 3, to have the number in the cell being divisible by 9. Note that, among all integers $x$ ranging from 1 to 2022, there are $\left\lfloor \frac{2\cdot 2022-1+3}{2\cdot 3} \right\rfloor = 674$ with $2x - 1$ being divisible by 3, and $\left\lfloor \frac{2\cdot 2022-1+9}{2\cdot 9} \right\rfloor = 225$ with $2x - 1$ being divisible by 9. The number of $(m, n)$ with one of $2m - 1$ and $2n - 1$ being divisible by 9, and the other not being divisible by 3, can be calculated as $225 \cdot (2022 - 674) \cdot 2 = 606600$. The number of $(m, n)$ with both $2m - 1$ and $2n - 1$ being divisible by 3 can be calculated as $674^2 = 454276$. Hence the answer is $606600 + 454276 = \boxed{1060876}$.

13. A positive divisor of $69^{420}$ must be in the form of $3^x 23^y$ where $x$ and $y$ are non-negative integers of at most 420. As $3^x 23^y \equiv 3^{x+y} \pmod{10}$, we have $x + y \equiv 3 \pmod 4$. As there are $106, 105, 105$ and $105$ non-negative integers of at most 420 which has a remainder of $0, 1, 2$ and $3$ when divided by 4, so our answer is $106 \cdot 105 + 105 \cdot 105 + 105 \cdot 105 + 105 \cdot 106 = \boxed{44310}$

14. Let us use the figure below as an aid.




Let the red ones be at the $p_1 < p_2 < p_3 < p_4$-th position. Then let

$$a = p_1 - 1, b = p_2 - p_1 - 1, c = p_3 - p_2 - 1, d = p_4 - p_3 - 1, e = 16 - p_4 - 1$$

be the number of blue integers on the left of the first red integer, between the first and the second red integers, ..., on the right of the fourth red integer respectively. Note that they sum up to 12. Let us denote an area be a set of blue integers which are pairwise adjacent. However, the blue integers in an area must not be put into another area, because the red integer immediately in the right (or in the left) is longer (or shorter), but is put in the left (or right) of the permuted blue integer, which leads to a contradiction. Therefore, a blue integer may only permute within its area. Then we would have

$$n \geq a!\ b!\ c!\ d!\ e!.$$

Note that we have $a!\ b! \geq \left\lfloor \frac{a+b}{2} \right\rfloor! \left\lceil \frac{a+b}{2} \right\rceil!$. By repeatedly applying this algorithm, we should achieve the minimum when $(a, b, c, d, e) = (2, 2, 2, 3, 3)$. Hence the minimum of $n$ would be $2 \times 2 \times 2 \times 6 \times 6 = \boxed{288}$.

15. Answer is $\boxed{15}$, will type solution later.

16.

$$\begin{aligned} abcd + \frac{1}{abcd} &= \left(a+\frac{1}{b}\right)\left(b+\frac{1}{c}\right)\left(c+\frac{1}{d}\right)\left(d+\frac{1}{a}\right) - \sum_{cyc}\left(a+\frac{1}{b}\right)\left(b+\frac{1}{c}\right) + 2 \\ &= 2\cdot 3\cdot 4\cdot 5 - (2\cdot 3 + 3\cdot 4 + 4\cdot 5 + 5\cdot 2) + 2 \\ &= \boxed{74} \end{aligned}$$

17. Taking modulo 3, we have

$$-b^2 \equiv (-1)^n \pmod 3.$$

As a square must be congruent to 0 or 1 modulo 3, the LHS must be congruent to 0 or $-1$. However, the RHS could only be congruent to 1 or $-1$, so $n$ must be odd. Indeed, if $n$ is odd, $a = b = 10(200)^{(n-1)/2}$ satisfies the condition. Hence the sum of all such $n$ is

$$1 + 3 + \cdots + 99 = 50^2 = \boxed{2500}.$$

18. $\angle FCG = 45^\circ = \angle DBC = \angle FEC$, so $\triangle GFC \sim ]triangle CFE$. Hence $CF^2 = FG\cdot EF = 25(25+7)$ i.e. $CF = 20\sqrt{2}$. Answer is $\boxed{10003}$, will type solution later.

19. Answer is $\boxed{1001001}$, will type solution later.